Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Using the Sensor Overview 6.5.0

Learn more at Stellar Cyber Academy.

The following links take you to courses on the Stellar Cyber Academy technical training portal where you can learn more about a topic by watching the suggested lessons.

M03 Modular Sensor Concepts and Configuration (08h:56m)

Modular Sensors Introduction (10m:00s)

Discover how Modular Sensors integrate into the Stellar Cyber Platform architecture. Learn about different types of Modular Sensors, their deployment in virtual and hardware forms, and their roles in data collection and analysis.

M06 Open XDR Linux Server Sensor Installation and Configuration (01h:21m)

Server Sensor Authorization (15m:00s)

Learn how to authorize the Linux Server Sensor within the Stellar Cyber Platform, then follow a practical demonstration of the complete authorization workflow in the data processor. Explore how to configure sensor profiles to enable secure data transfer, assign custom settings for tailored data monitoring, and confirm sensor status to establish real-time data flow into the data lake.

The first time you access a link on the portal during a session, you must log in to access content.

The **System | DATA SOURCE MANAGEMENT | Sensors | Sensors** page provides a list of all known sensors that are connected to the Stellar Cyber Platform.

When a sensor is installed, it is configured with the IP address of the Stellar Cyber Platform. When the sensor comes up, it connects to the Stellar Cyber Platform and appears in the Sensor table automatically.

The Sensor table provides the following functions:

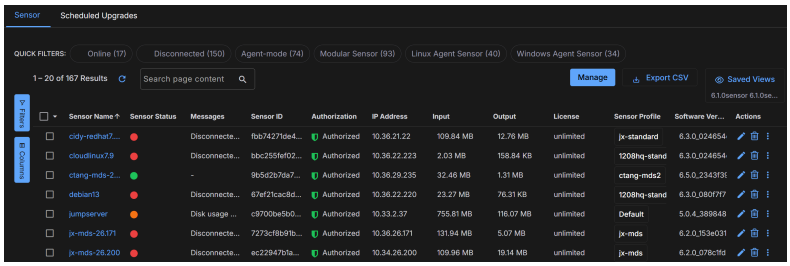
- List the known sensors with basic information such as host name, sensor ID, IP address, and software version

- View information such as operational status, authorization, and traffic counters
- Use Quick Filters to focus the display temporarily on just those sensors you need to see (for example, all Windows Server Sensors)
- Configure sensors with the use of Sensor Profiles
- Remote upgrade of sensors
- Remote downgrades of sensors (rollbacks)
- Remove a sensor from the system
- Launch the CLI for sensors.
- Apply CA certificates to sensors.
- Assign primary and secondary aggregators to sensors.
- Download sensor logs for troubleshooting

From here, you can also navigate to the Scheduled Upgrades tab to keep track of pending updates for sensors.

Sensor Overview List

When the Sensor Overview page appears, it shows a table of known sensors. The table is scrollable and provides standard table functionality. In addition, you can click the **Sensor Name** for any listed sensor to drill to its Sensor Details page, with detailed performance metrics and configuration information. You can also use the Quick Filters at the top of the display to focus the display temporarily on just those sensors you need to see (for example, all disconnected servers).

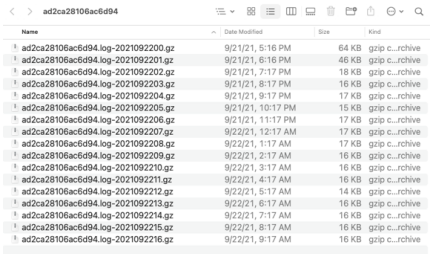


Each sensor is listed in the table with its own row. The columns include:

- **[ ]** – Check box to select sensors to Upgrade, open CLI Access, or Manage. Buttons for these features appear above the table appear when one or more sensors are selected.
- **Name** – The host name where the sensor is located. This value may not be unique within the system since it depends on the configuration done on the system where the sensor runs. Click the name to see sensor details.
- **Sensor Status** – The current status of the sensor is indicated with a color-coded LED. Refer to **About the Sensor Status LEDs** for details on the different conditions each of the sensor status LEDs can indicate.
- **Messages** – Status message, if any, for the sensor.
- **Internal Sensor ID** – A unique ID of the sensor, as assigned by the Data Processor.
- **Sensor ID** – A unique ID of the sensor, as configured by the user.
- **Authorization** – The authorization status of the sensor.
- **IP Address** – The current IP address of the sensor in use. Communication from the Data Processor to the sensor will use this address.

- **Traffic Input** – Total number bytes the Sensor has received. This represents the raw data received by the sensor.
- **Traffic Metadata Output** – Total number of bytes sent from the Sensor to the DP. This is the amount of data sent after the raw data received has been enriched, deduplicated, and converted to Interflow.
- **License** – The current license type.
- **Security Sensor License** – Indicates whether the sensor is either a physical (device) Security Sensor, or a Modular Sensor with Security Sensor features enabled in its Modular Sensor Profile (IDS and/or Malware Sandbox).

  Adding this column is a handy way to keep track of which sensors are consuming your purchased Security Sensor licenses.
- **Sensor Profile** – The sensor profile that is applied to the sensor. A sensor profile contains configuration information that is shared between any number of sensors. See the Sensor Profiles page for more information.
- **Software Version** – The version of software that is running.
- **Log Forwarder** – Log forwarding is enabled.
- **Edit** – Edit the settings for a selected sensor. Details are below.....
- **Delete** – Discard the selected sensor.
- **More options** - Click this ⋮ button to display a menu option to **Download Debug Logs**. Select this menu to download a gzip of the debug logs available for that sensor. The download is saved according to your browser settings for download files. The unpacked zip will appear similar to this example:

  You can also use this menu to recollect the latest debug logs for the sensor before downloading them.

Additional columns, not displayed by default, are available.

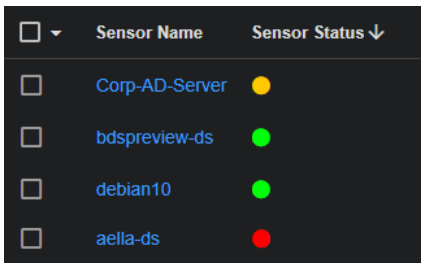
- APT
- Feature
- Feedback
- IDS
- Last Seen
- Location

- Log Forwarder
- Malware Analyze Res
- Mode
- Need Upgrade
- Note
- OS
- Packet Forwarding Interface
- Platform (for example, **Virtual_Machine**)
- Reduction Ratio
- SSH Tunnel
- Tenant ID
- Tenant Name
- Timezone
- Tunnel info
- Upgrade Result
- VM Name

See the Tables page for more information on working with Tables.

About the Sensor Status LEDs

The **Sensor Status** column displays color-coded LEDs indicating the health of each sensor at a glance:



- **Green** – The sensor is operating normally and there are no danger or warning conditions detected.
- **Yellow** – At least one warning condition is affecting the sensor.

  Click here to see the possible warning conditions:

  - Sensor disk usage exceeds 90%.

    Windows Server Sensors check the usage of the system disk every 15 seconds using the Python **psutil** library. If the disk usage is over 90%, the system displays the Disk usage more than 90% warning condition.

  - Sensor CPU usage exceeds 99%.
  - Sensor feedback contains a `config mismatch` error message.

    - Sensor has not updated its status in at least five minutes.
    - Sensor feedback shows a failure to apply a profile with the Tenable Nessus feature enabled to a sensor running version 4.3.3 or lower. (Tenable Nessus support was introduced in 4.3.4.)
    - Sensor feedback contains at least one `warning` message .
- **Red** – At least one danger condition is affecting the sensor.

  Click there to see the possible danger conditions:

    - Sensor has an invalid Engine ID (`engid`)
    - Sensor has a conflicting Engine ID (`engid`)
    - Sensor configuration is out of sync (either the `ver_sent` does not match received `ver_ack` or one of the two messages is missing)
    - Sensor memory usage exceeds 95%
    - Sensor feedback contains a `config failure` message
    - Sensor feedback shows at least one failure during the application of a new sensor profile.

      The exception to this is a failure to apply a modular sensor profile with the Tenable Nessus feature enabled to a sensor running 4.3.3 or lower. Those failures are reported as warnings and only display the yellow LED, as described above.

Using Quick Filters in the Sensor List 6.5.0

The Sensor List includes a handy set of Quick Filters at the top of the table that let you temporarily focus the display on just those sensors you want to see for your current workflow. Selecting one of the Quick Filters automatically limits the display to just those sensors matching the filter (for example, all connected sensors).

- Remove an applied Quick Filter by selecting it again, returning the display to an unfiltered state.
- Apply a new Quick Filter to see a different set of sensors. Quick Filters are not cumulative; only one is applied at a time.

The following Quick Filters are available:

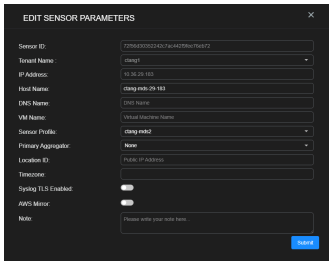
- **Online** – Shows only those sensors that are currently online and connected to the Stellar Cyber Platform.
- **Disconnected** – Shows only those sensors that are currently disconnected from the Stellar Cyber Platform.
- **Agent-mode** – Shows all Server Sensors (Linux and Windows). Server Sensors are sometimes referred to as "agents," because they are installed co-resident on a monitored host.
- **Modular Sensor** – Shows all sensors with a modular sensor profile applied.

- **Linux Agent Sensor** – Shows all Linux Server Sensors.
- **Windows Agent Sens** – Shows all Windows Server Sensors.
- **Device-mode** – Shows all device sensors. Device sensors include all sensors that are not Server Sensors
- **Security Sensor** – Shows all Security Sensors. Security Sensors are a legacy sensor type with IDS and malware sandboxing enabled. These features are now enabled in a Modular Sensor profile.

Refer to **Sensors** for a discussion of the different types of sensors used in a Stellar Cyber deployment.

Editing a Sensor's Settings

Some of the information associated with each sensor can be edited by clicking the button at the right side of a sensor entry in the table:



Most fields are read-only. The available fields include:

- **Sensor ID** – The internal ID for the sensor.
- **Tenant Name** – The tenant for the sensor.

  Note that if you change a sensor's tenant assignment, any CA certificates assigned to the sensor are automatically cleared without any notification in the user interface.

- **IP Address** – The IP address of the sensor.
- **Host Name** – The host name of the sensor. This value will be used in the first column of the sensor list described above. This field does not support multibyte characters.
- **DNS Name** – The DNS Name of the sensor. This field does not support multibyte characters.
- **VM Name** – The name of the VM in which the sensor is running.
- **Sensor Profile** – This field is a dropdown menu of the available sensor profiles.
- **Primary Aggregator** – This field is a dropdown menu of the available data aggregators. Once you assign a Primary Aggregator, a new **Secondary Aggregator** field appears, allowing to specify the aggregator to use in situations where the primary aggregator is unreachable.

  You can also assign aggregators from directly within the Sensor Overview. Start by checking the box for the sensor to which you want to assign an aggregator. Then,

click the **Assign Aggregator** button that appears at the top of the table to open a dialog box where you can assign both a primary and secondary aggregator. See **Assigning Aggregators to Sensors** for details.

- **Location ID** – The gateway IP address. Stellar Cyber automatically detects the gateway IP address, but this lets you set it manually.
- **Timezone** – The timezone in which the sensor is operating.

  You can only change the timezone for device sensors. The **Timezone** option is hidden for Server Sensors.

  See **Timezones and Log Ingestion** for information on how the sensor's timezone is used during log ingestion.
- **PKT Forwarding Interface** – The VXLAN interface number.
- **Syslog TLS Enabled** – Enable to ingest TLS-encrypted logs. See Ingesting Logs over TLS for more information. In general:
  - Click the adjacent **CA Certificate** button to download the self-signed certificate from the sensor for installation on your log source.
  - If your log source does not support the self-signed certificate, you can apply a commercial server certificate to the sensor for TLS log ingestion by first uploading the certificate to Stellar Cyber in the Certificates page and then applying it to the sensor on this page using the **Apply Certificate | Apply Certificate to Log Forwarder** button.

  Disabling the **Syslog TLS Enabled** option for a Modular Sensor that is using a server certificate removes the association between the sensor and the certificate. If you then re-enable **Syslog TLS Enabled**, the sensor reverts to using the self-signed certificate instead of using the server certificate that was previously configured. If this happens, you can just reapply the server certificate using the the **Apply Certificate | Apply Certificate to Log Forwarder** button. 5.3.0
- **AWS Mirror** – Enable this to monitor traffic from a mirror port you configured on AWS. See Configuring AWS Port Mirroring for details.

  If you enable **AWS Mirror**, the following additional fields appear:
  - **Physical Ethernet Port** – This is the index number of the physical interface the UDP traffic arrives on.
  - **Port** – The UDP port for the traffic.
  - **VNI** – This is the VXLAN ID you configured on AWS.
- **Note** – Use this to keep notes about the sensor. This field does not support multibyte characters.

To make changes, edit the fields as desired and click **SUBMIT** to save.

Timezones and Log Ingestion

Stellar Cyber converts all log timestamps to UTC during log ingestion. Timezones are handled as follows:

- If a log **includes** the timezone, Stellar Cyber preserves that time setting during the conversion to UTC.
- If a log **does not include** a timezone, Stellar Cyber uses the timezone of the receiving sensor.

During installation, the timezone for sensors are automatically set to UTC+0. Since the logs for some security products might only include the local time without a timezone, Stellar Cyber recommends that you set the sensor timezone to the same timezone as your security product.

Refer to **Best Practices for NTP and Timezones** for more information.

Deleting a Sensor

You can click the button at the right of a sensor's entry in the table to remove it from the list. If you delete the sensor, the sensor is removed from the list, and the sensor deletes its configuration so it will not be discovered by Stellar Cyber.

Removing (Uninstalling) a Server Sensor 6.5.0

You can select one or more Server Sensors and click the **Remove** button to uninstall them from their host(s). When you remove a Server Sensor from the Sensor List, Stellar Cyber uses a managed uninstall workflow instead of deleting the sensor record immediately.

After you click **Remove**, the sensor enters a **Deleting** status while Stellar Cyber waits for the uninstall result. The uninstall request is tracked in the **Task Management** page under the sensor uninstall task.

- If the uninstall succeeds, Stellar Cyber removes the sensor from the list automatically. If the uninstall fails, the sensor shows an error status with the message **Sensor uninstallation failed.**
- If the uninstall fails, you can choose one of the following actions:
  - **Retry** to start another uninstall attempt
  - **Delete Anyway** to remove the sensor record without attempting another uninstall

This workflow gives you visibility into sensor removal progress and lets you confirm whether the uninstall completed successfully before the sensor record is removed.

Upgrading Sensor Software

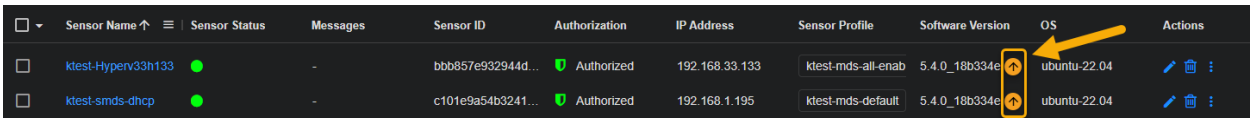
Sensors can be upgraded manually:

- **Manual Upgrades** – You can use the instructions in this section to upgrade sensors manually, selecting one or more sensors in the Sensor List and clicking the **Upgrade Software** button.

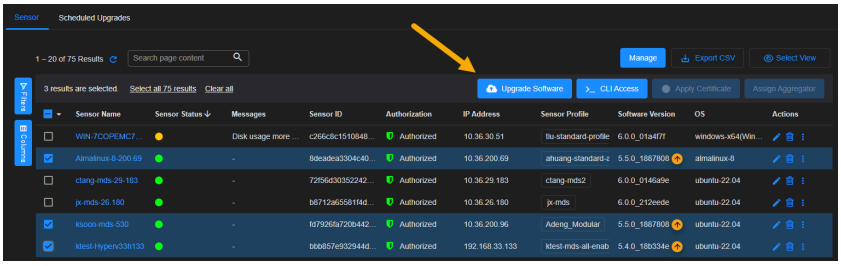
Use the following procedure to upgrade sensors manually:

1. Navigate to the **System | DATA SOURCE MANAGEMENT | Sensors | Sensors** list and inspect the Software Version column. Sensors that have an upgrade available are listed with a distinctive icon in this column, as shown below:

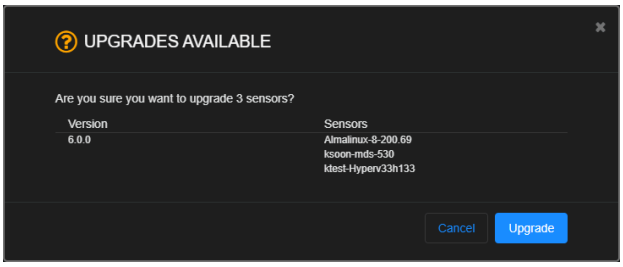
   The system also automatically identifies sensors that are eligible for an upgrade and informs you with a banner showing the total and including an **Upgrade Now** button you can use to streamline the process.



2. Check the boxes of the sensors you want to upgrade in the Sensor List and click the **Upgrade Software** button at the top of the list.

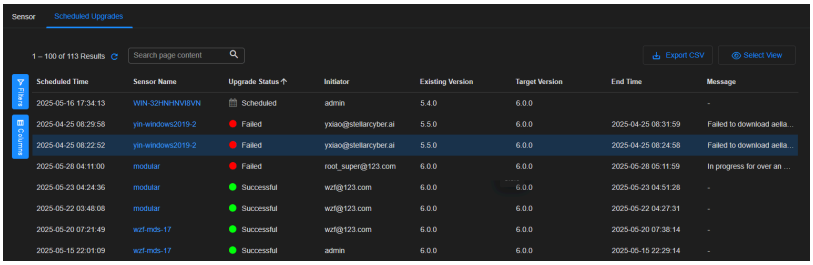


3. Stellar Cyber displays a confirmation prompt listing the sensors you've selected for the upgrade. Verify the sensors and click the **Upgrade** button, as illustrated below.



4. Stellar Cyber evaluates the sensors selected for the upgrade. If one of the sensors you selected for the upgrade already has been scheduled for an upgrade, the system informs you of the situation and queues the remaining sensors for an upgrade.

5. Once the sensors are scheduled for an upgrade, you can keep track of the progress in the **Scheduled Upgrades** tab. The illustration below shows an example of the Scheduled Upgrades tab:



What Happens Next?

Once a sensor is selected for upgrade, behind the scenes, Stellar Cyber identifies the correct software version to apply and schedules the upgrade to take place during a period of low activity. If the sensor was offline when the upgrade was requested, Stellar Cyber continues to

Note that when you upgrade a sensor from the Sensor Overview page, all configuration settings are preserved acr he upgrade, including the DP's IP address, aggregator addresses, tenant ID, log level settings, and DHCP server log settings.

Downgrading Sensor Software (Rollbacks)

This section describes a feature that is currently only available as part of an Early Access Program and may not be enabled in your version of the Stellar Cyber Platform. Contact your sales representative for information on enabling this feature and trying it out with Stellar Cyber's Customer Success team.

You can downgrade sensor software from the Sensor List. This may be useful in situations where an upgrade produces unexpected results or when working with Stellar Cyber Customer Success to troubleshoot an issue.

Keep in mind the following when downgrading sensor software:

- Downgrades are supported for Windows Server Sensors, Linux Server Sensors, and Modular Sensors.
- Downgrades are only supported from version 6.0.0 to version 5.5.0.
- When you launch the Software Downgrade interface, it lists only those sensors that are currently eligible for a downgrade.

Use the following procedure to downgrade sensor software:

1. Select the **Manage | Software Downgrade** option. A dialog box similar to the following appears:



2. Select the version for the downgrade from the **Available Software** list.

   The only supported downgrade path in this release is from 6.0.0 to 5.5.0.

3. Select the sensors to downgrade from the **Target Sensors** list. Only sensors supported for downgrade to the selected version are listed.
4. Click **Submit**. The downgrade begins.

   The downgrade takes approximately five minutes. No logs are collected by the sensor while the downgrade takes places. Log collection starts again automatically once the downgrade concludes.

5. Refresh the Sensor table to see the results. If you do not see the **Upgrade Result**column:

   a. Click the **Columns** tab available at the left of the Sensor List.

b. Scroll to the bottom of the list.

c. Check the bo **Upgrade Result**. The column is immediately added to the table.

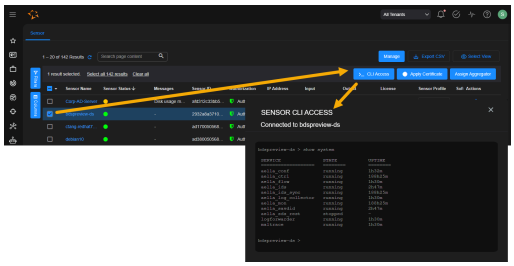
> Even though the column is named **Upgrade Result**, it provides information on both upgrade and downgrade operations.

d. Click the **Result** button in the **Upgrade Result** column to see information on the success or failure of the operation.

> Note that when you downgrade a sensor, all configuration settings still available in the downgraded version are preserved across the transition, including the CM IP address, aggregator addresses, tenant ID, log level settings, and DHCP server log settings. However, any settings that were only available in the newer software version will not be available in the downgraded version.

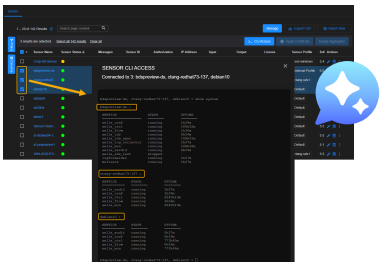
Using the Sensor CLI 6.2.0

You can select one or more sensors in the Sensor Overview table and click the **>_CLI Access** button to open a command line session with the corresponding sensor(s) in the Sensor CLI Access window, as illustrated in the figure below:



Note the following:

- The available CLI commands are a superset of the CLI commands available when connected directly to a sensor. Refer to **Available Commands in the Sensor CLI Access Window** for a summary of the available commands.
- You can only use the CLI Access feature with sensors that are connected and running a minimum version of 4.2.x. Sensors do not need to be authorized for CLI access.
- You can use your browser's right-click context menu to copy text from the Sensor CLI Access window and paste it for use elsewhere. The window is scrollable, too, so you can see the output of commands that do not fit entirely in the window.
- You can select multiple sensors before clicking the **>_CLI Access** button. When you use this technique, the CLI issues any command to all connected sensors and returns the results under a heading indicating each sensor's name. For example, the illustration below show the results of a **show system** command issued to three separate sensors simultaneously.



Sensor CLI Access Commands

The following types of commands are available in the Sensor CLI Access window:

- **show** – See the settings and statuses for different options on the sensor.
- **set** – Enable and configure options on the sensor.
- **unset** – Disable options on the sensor.
- **copy** – Copy specified trace files or support files.
- **exec** – Execute a specified JSON task or a tcpreset-stop action.

You can use the ? or help parameter with any of these commands to see the available commands of the corresponding type. For example:

- **set ?** or **set help**

You can also use the ? with a specific command to see its available arguments. For example:

```
mysensor > set ntp ?
<NTP server> Specify NTP server name or IP address
```

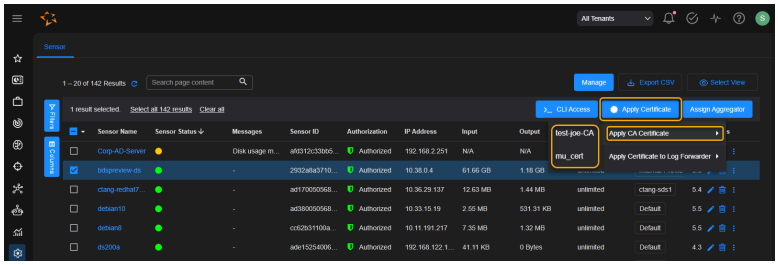
The Sensor CLI Access window supports copy-and-paste functionality. 5.4.0

Refer to **Available Commands in the Sensor CLI Access Window** for a summary of the available commands.

Applying CA Certificates to Sensors

You can select one or more sensors in the Sensor Overview table and click the **Apply Certificate | Apply CA Certificate** button to assign a CA certificate to them. In response, a dropdown appears listing the CA certificates you've added to Stellar Cyber in System | Saved Objects | Certificates that match the selected sensor's tenant, allowing you to click the one you want to apply.

Prior to the 5.1.1 release, the button was just named **Apply CA Certificate**.



If you are assigning the same CA certificate to multiple sensors belonging to the same tenant, you can use the **Change Columns** button to add the **Tenant Name** column to the display, click in the column heading to sort on it, and then select all of the tenant's sensors to which you want to apply the CA certificate.

You only need to apply CA certificates to those sensors that sit behind firewalls using an SSL inspection service. Without the CA certificate from the SSL inspection service applied, the sensor cannot retrieve the necessary package for upgrading. Make sure you choose the CA certificate used by the SSL inspection service on the firewall protecting the sensor(s).

Note that it is possible that your sensors may be behind a different firewall than the DP. Make sure you assign the CA certificate from the correct firewall to the sensor(s).

DPs behind firewalls using an SSL inspection service also require a CA certificate. You assign certificates to DPs using the **CA Certificate** entry in the **Data Processor** section of System | ORGANIZATION MANAGEMENT | Settings.

If a certificate you expect to see doesn't appear when you select the **Apply CA Certificate** option, make sure you uploaded it with the correct tenant assignment in **System | Saved Objects | Certificates**.

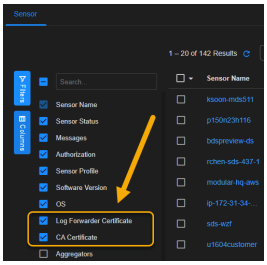
Automatic Handling of MS-Windows Carriage Returns in Certificate and Key Files 6.6.0

Stellar Cyber automatically handles certificates and key files that include the MS-Windows carriage return character (`\r`, or `0x0D` in hexadecimal). You do not need to remove them manually.

Removing a Certificate from a Sensor

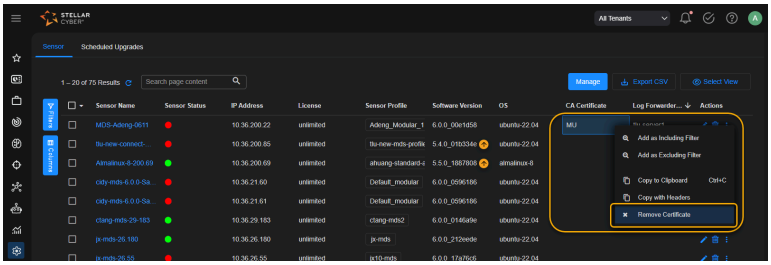
You can also remove an applied certificate from a Sensor in the Sensor Overview table. The procedure is similar for both CA Certificates and Log Forwarder certificates.

1. Start by adding a column to the Sensor List of the type corresponding to the type of certificate you want to remove. Click the **Columns** entry at the left of the table and select one or both of the following, as illustrated below:
    - **Log Forwarder Certificate**
    - **CA Certificate**



2. Locate the sensor whose certificate you want to remove and right-click its entry in either the **Log Forwarder Certificate** or **CA Certificate** column. Then, select the **Remove**

Certificate option from the context menu that appears, as illustrated below. Keep in mind that you must remove all associations from a certificate before you can delete it in System | Saved Objects | Certificates.



Assigning Aggregators to Sensors

You can assign an aggregators to sensors from directly within the Sensor Overview by selecting their entries in the list and clicking the **Assign Aggregator** button at the top of the list.

An aggregator is a Modular Sensor with the **Aggregator** feature enabled in its Modular Sensor Profile. An aggregator forwards aggregated traffic received from other Stellar Cyber sensors to the Stellar Cyber Platform configured in its modular sensor profile.

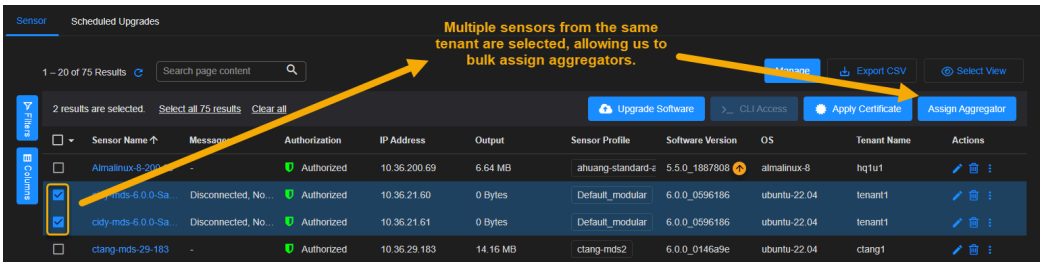
You can also assign an aggregator to a sensor from the sensor's Edit Sensor dialog box, just as you have in previous releases. The advantage of assigning aggregators directly from the Sensor Overview is that you can assign the same aggregators to multiple sensors in one step (a bulk edit, in other words). You also don't need to open any extra dialog boxes when you assign aggregators from the Sensor Overview – it all happens right there in the table.

Use the following procedure to assign aggregators to sensors from within the Sensor Overview:

1. Check the boxes for the sensors to which you want to assign primary and secondary aggregators.

   The **Assign Aggregator** button appears with the other bulk edit controls at the top of the Sensor Overview, as illustrated below.

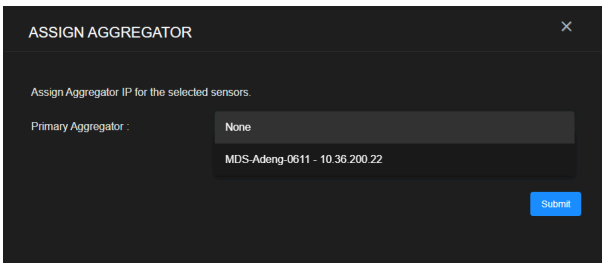
   If you select multiple sensors, they must all belong to the same tenant in order to be able to perform a bulk aggregator assignment. If the **Assign Aggregator** button appears but is grayed out, you may have selected sensors from different tenants.



2. Click the **Assign Aggregator** button to open the Assign Aggregator dialog box.
   - If all selected sensors are already assigned to the same aggregators, the dialog box shows the current assignments.
   - If any of the selected sensors are assigned to different aggregators (or are all unassigned), the dialog box appears without any aggregator assignments

displays, allowing you to configure them now from scratch.

3. Use the **Primary Aggregator** dropdown to select the primary aggregator for the selected sensors. The dropdown lists all available aggregators for the tenant to which the sensors belong.



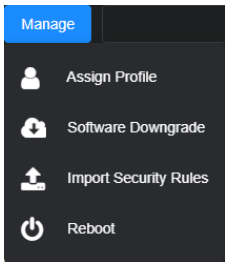
4. Once you have selected a Primary Aggregator, the Secondary Aggregator dropdown appears, allowing you to specify the aggregator to use in situations where the primary aggregator is unreachable. The dropdown lists all aggregators for the tenant to which the sensors belong, except for the one you just assigned as the Primary Aggregator.
5. When you have finished choosing the aggregators for the selected sensors, click **Submit**.

Keep in mind that TCP Port 8080 must be open between the sensor and the aggregator for successful communications.

To understand how Windows and Linux Server Sensors handle scheduled update retrievals when using an aggregator, see **Server Sensor Update Behavior through an Aggregator**.

Sensor Management Menu

The **Manage** button is located above Sensor Overview table. Clicking it displays the menu shown below. These options are all related to the management of the sensors and each option displays its own dialog box.



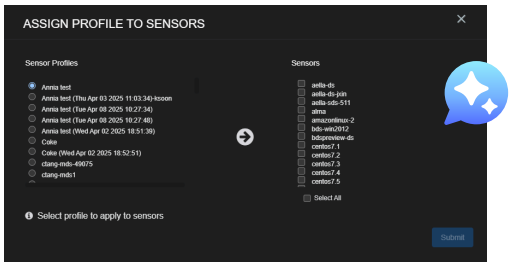
The following sections describe the available functions.

Sensor Assign Profile

All sensors must be assigned to a profile. Except for the **default** Profile, you create your own profiles. For more information see the Sensor Profiles page.

To assign profiles, select the **Manage | Assign Profile** option.

A dialog box similar to the following appears.



Use the Assign Profile to Sensors dialog box as follows:

1. Available profiles are listed in the **SENSOR PROFILES** list. Select one with the radio-button control.
2. Available sensors are listed in the **SENSORS** list. Select the sensor(s) to which you want to apply the profile selected in the **SENSOR PROFILES** list by checking their boxes.
3. Apply the selected profile to the selected sensor(s) by clicking the **Submit** button.

Changes in this dialog box do not take effect until the **Submit** button is clicked. If the X button is used to close the dialog box no changes are made.

Sensor Downgrade 6.0.0

You can use the **Manage | Downgrade Software** option to roll a sensor back to a specified version.

Refer to **Downgrading Sensor Software (Rollbacks)** for details on using this feature.

Sensor Import Security Rules

You can use the **Manage | Import Security Rules** option to import custom Suricata (IDS) security rules from an external file for use by sensors with the **ML-IDS** (Malware) feature enabled in their sensor profiles.

Use the following procedure to import custom security rules:

1. From the Sensor table, select **Manage** and then select **Import Security Rules**.

   A dialog box similar to the following appears:

2. Select a local file by using **Choose File**.

   See below for details on the rule file format.

3. Select the sensors that will use the security rules from the SENSORS list.
4. Select **Upload and Deploy** to apply the imported security rules to the selected sensors.

Rule File Format

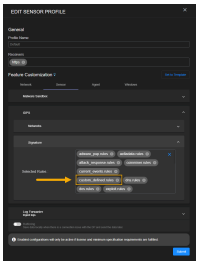
The rule file must follow standard Suricata rule format with each rule on a single line. Here is an example of a custom rule in the correct format:

```
alert tcp $HOME_NET any -> $EXTERNAL_NET any (msg:"ET TROJAN Likely Bot Nick in IRC (USA +..)"; flow:established,to_server; flowbits:isset,is_proto_irc; content:"NICK "; pcre:"/NICK .*USA.*[0-9]{3,}/i"; reference:url,doc.emergingthreats.net/2008124; classtype:trojan-activity; sid:2008124; rev:2;)
```

Note the following important points for custom rules:

- Because Stellar Cyber sensors with the **ML-IDS** feature enabled operate as an IDS rather than an IPS, the action for your security rules must be **alert**, as shown above.
- The values for `$HOME_NET` and `$EXTERNAL_NET` are specified in the Sensor Profile:
    - **Modular Sensor Profiles** – Feature Customization | IDS | Networks
- Once imported, custom rules are stored in the **custom_defined.rules** signature set. They are enabled in the sensor profiles for all sensors to which you applied them by default and can be seen in the Sensor Profile under **Signature | Selected Rules**, as illustrated below.

  You can toggle your custom rules on and off using the **custom_defined.rules** entry shown below, as well.



> Each time you import custom security rules to a sensor, the new rules completely overwrite any existing entries in the **custom_defined.rules** signature set. New rules are not appended to the existing rules.

Troubleshooting Custom Security Rules

- You can check the Sensor's **ids** feedback message to see the result of an upload and deploy. For example, here is the result of a successful upload and deploy:

```
"ids": {
    "msg": "Success",
    "status": true,
    "timestamp": 1690000900,
    "datetime": "2023-07-21 21:41:40"
  },
```

- You can keep track of IDS events and malware detections by using the `show maltrace` command from the Sensor's CLI. For example:

```
DataSensor> show maltrace
Maltrace Event:
Captured:                                   0
Truncated:                                  0
Prefiltered:                                0
Verdict Found in Local Cache:               0
Verdict Found in Remote Cache:              0
```

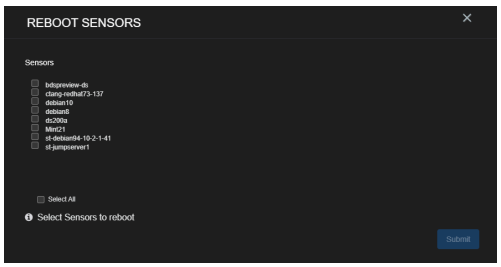
```
Passed to Local Scan Engine:                  0
Filtered on Local Scan Engine:                0
File Analyses Failure on Local Scan Engine:   0
Local Scan Engine Failure:                    0
Verdict Given by Local Scan Engine:           0
Uploaded to Sandbox:                          0
Failure Over Sandbox:                         0
Sandbox Scan Succeeded:                       0
Sandbox Timeout:                              0
Sent Result to Central Manager:               0

IDS Event:
Received:         67
Excluded:         0
Sent Result:      0
Unknown Score:    67

Output buffering:
Buffered:        67
Outputed:        66
```

Rebooting Sensors Remotely

Sensors can be rebooted remotely. Select the **Manage | Reboot** option to reboot one or more sensors remotely. A dialog box similar to the following appears:



Select the sensors to reboot by checking their boxes in the **SENSORS** list and then select **Submit** to start the reboot process on the selected sensors.

Rebooting a sensor does not reboot the host on which the sensor is installed.

Using the Scheduled Upgrades Tab

The Scheduled Upgrades tab tracks the success or failure of sensor upgrades you have launched.

For details on how Server Sensors retrieve updates and upgrades when an aggregator is configured, see **Server Sensor Update Behavior through an Aggregator**.

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